American Computer Science League

2021-2022 ● Contest 2: Shorts ● Classroom Division

1. Prefix/Infix/Postfix

Evaluate the following prefix expression if all numbers are single digits:

+ / 9 3 + * 4 − 8 3 ^ 5 2

A. 33
B. 42
C. 48
D. 52
E. 63

2. Prefix/Infix/Postfix

Convert this infix expression to postfix:

$$\frac{\left(A^2 + B\right)\left(B^2 - C\right)}{B + D^2}$$

A. A 2 B ^ + B 2 ^ C − * B D 2 ^ + /
B. A 2 ^ B + B 2 ^ C − * B D 2 ^ / +
C. A 2 ^ B + B 2 ^ C − * D B 2 ^ + /
D. A 2 ^ B + B 2 ^ C − * B D 2 ^ + /
E. A 2 ^ B + B ^ 2 C − * B D 2 ^ + /

3. Bit-String Flicking

Evaluate:

01100 OR 11011 AND 01010

A. 11011
B. 01110
C. 01010
D. 10011
E. 11101

4. Bit-String Flicking

Evaluate:

(LCIRC-2 01100) AND NOT 11010 OR (RSHIFT-1 01100)

A. 11010
B. 10101
C. 00111
D. 10110
E. 01010

5. LISP

Evaluate the following LISP expression:

(CAADDR '((1 2 (3)) (4 (5) 6) ((5 4) (3 (2)))))

A. ((5 4))
B. (3 (2))
C. (1 2 (3))
D. (5 4)
E. 5

6. Prefix/Infix/Postfix

Evaluate the following postfix expression if all numbers are single digits:

6 2 ^ 6 − 5 / 8 6 2 + / − 5 /

A. 0
B. 1
C. 2
D. 3
E. 4

7. Prefix/Infix/Postfix

Convert this prefix expression into postfix if all numbers are single digits:

/ * + − 5 4 5 ^ 1 4 / 9 3

A. 5 4 5 − + 1 4 ^ * 9 3 / /
B. 5 4 − + 5 1 4 ^ * 9 3 / /
C. 5 4 − 5 + 1 4 * ^ 9 3 / /
D. 5 4 − 5 + 1 4 ^ / 9 3 * /
E. 5 4 − 5 + 1 4 ^ * 9 3 / /

8. Bit-String Flicking

Evaluate the following:

00100 OR 10100 AND NOT (01111 OR 00101)

A. 10100
B. 00100
C. 10000
D. 01111
E. 00101

9. Bit-String Flicking Evaluate the following expression: (LCIRC-2 (10011 AND (NOT 11101) OR (RSHIFT-2 11010)))	A. 00110 B. 10011 C. 11000 D. 00010 E. 11001
10. LISP Evaluate the following LISP expression: (CAR (CDR '((0 1) (3 2 1) (2 4 1) (3 2))))	A. (0 1) B. 3 C. (3 2 1) D. ((2 4 1) (3 2)) E. (2 4 1)